



Modelling Financial Markets

Seminar 0: Financial Markets in 2026 — Hands-on



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Acronyms used in this material

- ▣ AI: Artificial Intelligence
- ▣ API: Application Programming Interface
- ▣ AR: AutoRegressive (model)
- ▣ BET: Bucharest Exchange Trading (index)
- ▣ BET-TR: BET Total Return (index)
- ▣ BNR: Banca Națională a României (the National Bank of Romania)
- ▣ BVB: Bursa de Valori București (the Bucharest Stock Exchange)
- ▣ CAGR: Compound Annual Growth Rate
- ▣ CAR: Cumulative Abnormal Return
- ▣ DCC: Dynamic Conditional Correlation
- ▣ ETF: Exchange-Traded Fund
- ▣ FRED: Federal Reserve Economic Data
- ▣ FX: Foreign eXchange
- ▣ HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ MAD: Median Absolute Deviation
- ▣ OLS: Ordinary Least Squares
- ▣ SE: Standard Error
- ▣ SR: Sharpe Ratio
- ▣ US: United States
- ▣ UTC: Coordinated Universal Time (Temps Universel Coordonné)

Outline

Objectives and Setup

Definitions Used Today

Getting and Cleaning Data

Part A: Pen-and-Paper Exercises

Part B: Computational Exercises

Part C: Open Question

Team Project

Seminar Objectives

By the end of this seminar you will be able to

- ▣ load daily market data and public online data through one function, and clean them
- ▣ derive when annualisation rules and correlation estimates fail
- ▣ attach standard errors and confidence intervals to returns, Sharpe ratios and correlations
- ▣ test whether a change is real: Fisher z, bootstrap, a first event study
- ▣ set up the GitHub repository of your team project

- ▣ Notebook for today: [Open the seminar notebook in Colab](#)
- ▣ Seminar: Antoaneta Amza, PhD candidate, antoaneta.amza@csie.ase.ro

Setup: Colab, Python and the Repository

```
# in a terminal (or a Colab cell with !)  
git clone https://github.com/danpele/MFM.git  
cd MFM  
ls data/market | head      # one CSV per symbol  
  
# Python packages used in Chapter 0  
pip install pandas numpy matplotlib scipy
```

- Colab needs no installation: open the notebook link and run all cells
- data/market: daily files with open, high, low, close, adjusted close, volume
- Quantlets/Ch_00: the code behind every chart of the lecture
- notebooks/EN, notebooks/R0: lecture and seminar notebooks

GitHub Workflow for the Team Project

One repository per team

- One member creates a public repository and invites the others
- README.md: research question, data, how to run
- One folder per result, Quantlet style: code + Metainfo.txt + chart

□ Example of the Quantlet format: [Quantlets/Ch_00/MFM_ch0_cross_asset](#)

Daily routine

- `git pull` before you start working
- Small commits with clear messages
- `git push` at the end of each session
- Never commit passwords or API keys

How This Seminar Works

- ▣ Two kinds of problems in every part
 - ▶ **[Solved]**: complete worked solutions, in the slides and in the notebook (code and results); use them as models
 - ▶ **[Proposed]**: you solve them; each one names its solved model; solutions are discussed in class
- ▣ **Part A**: pen-and-paper (A1–A4 warm-up, A5–A9 derivations); **Part B**: verification and inference on real data (Ex1–Ex8)
 - ▶ not “compute and compare with the lecture”, but “how sure are we, and when does the method fail?”
 - ▶ every exercise ends with interpretation questions that a number alone cannot answer
- ▣ **Part C**: an open question with no single method, a possible start for your team project (C1), an AI answer to critique (C2), and a research seed that replicates the lecture case study (C3)

Solved and Proposed Problems

Solved (models)

- A1–A4: warm-up (returns, drawdown, annualisation, CAGR)
- A5: volatility drag (derivation)
- Ex1: cross-asset dashboard; Ex1 Extended: standard errors and bootstrap
- Ex3: correlations and calendars
- Ex6: finding bad ticks
- Ex7: event study, Banca Transilvania

Proposed (model in brackets)

- A6 (A1–A4, A5), A7 (A5, Ex3), A8 Sharpe SE (A5, Ex1 Extended), A9 asynchronous closes (A5, Ex3)
- Ex2 drawdowns (A1–A4, Ex1); Ex2 Extended: is the drawdown extreme? (Ex1 Extended)
- Ex3 Extended: has the correlation changed? (Ex1 Extended)
- Ex4 stablecoins (A1–A4, Ex1); Ex4 Extended: structural break (Ex7)
- Ex5 BVB in EUR (Ex1); Ex5 Extended: is the currency effect significant? (Ex1 Extended, Ex7)
- Ex6 Extended (Ex6); Ex8 BRD (Ex7); C1 (Ex3); C2 AI critique (A1–A4, Ex1); C3 passive investing and demand elasticity (Ex1 Extended)

Definitions Used Today (1/5): Returns, Growth, Sharpe Ratio

- ▣ Simple return $R_t = P_t/P_{t-1} - 1$; log return $r_t = \ln(P_t/P_{t-1})$, with P_t the closing price on day t
- ▣ **Adjusted close**: price corrected for dividends and splits, used for stocks and ETFs
- ▣ **CAGR** (compound annual growth rate): $(P_{\text{end}}/P_{\text{start}})^{1/Y} - 1$, $Y = \text{calendar days} / 365.25$
 - ▶ $\text{CAGR} = e^{\mu_{\log}} - 1$, with $\mu_{\log}$ the annualised mean of log returns (geometric growth)
- ▣ **Arithmetic mean** μ_{arit} : annualised mean of simple returns; **growth multiple**: $P_{\text{end}}/P_{\text{start}}$
- ▣ **Sharpe ratio** (SR): $\sqrt{q} \bar{R}/s_R$, the mean over the standard deviation of daily simple returns, with the risk-free rate $r_f = 0$
 - ▶ $q = \text{observed trading days per year}$: about 252 for equities, 365 for crypto, about 260 for spot gold (weekday quotes only)
- ▣ **Year-on-year (YoY) growth**: $X_{\text{year end}}/X_{\text{previous year end}} - 1$

Definitions Used Today (2/5): Risk

- ▣ **Daily volatility:** the standard deviation $\hat{\sigma}_d$ of daily log returns
- ▣ **Annualised volatility:** $\hat{\sigma}_d\sqrt{q}$ (the " $\sqrt{P}$ rule"), exact only for uncorrelated returns (Exercise A6)
- ▣ **Autocorrelation** $\rho_j = \text{Corr}(r_t, r_{t-j})$: how today's return is related to the return j days earlier
- ▣ **Running peak:** $M_t = \max_{s \leq t} P_s$; **drawdown:** $DD_t = P_t/M_t - 1$
- ▣ **Maximum drawdown:** $\min_t DD_t$, the worst loss from a peak
- ▣ **Drawdown spell:** starts on the peak date and ends on the first close at or above that peak (full recovery)
- ▣ **Peak-to-trough decline:** trough/peak $- 1$ for a given episode

Definitions Used Today (3/5): Co-Movement and Data Terms

- **Correlation:** $\rho = \text{Cov}(r^X, r^Y) / (\sigma_X \sigma_Y) \in [-1, 1]$, computed on returns, never on prices
- Two estimators that answer different questions
 - ▶ **correlation over the whole period:** one number for all days in the period
 - ▶ **mean of the 252-day rolling correlation:** average of one-year windows (used in the lecture chart)
- Three ways to align two calendars
 - ▶ **common days:** join the *prices* on the days both assets trade, then compute returns (Friday $\rightarrow$ Monday for both)
 - ▶ **forward-fill:** repeat the last equity price on weekends, so its weekend returns are zero
 - ▶ **weekly returns:** Friday close to Friday close for both assets
- **Bad tick:** a wrong price in a data file, usually a spike reversed the next day
- **Rolling median:** the median of a moving window, e.g. 11 days centred on day t ; robust to single outliers
- **pp:** percentage point, the unit for differences between two rates in %

Definitions Used Today (4/5): Inference Tools

- **Standard error (SE)**: the standard deviation of an estimator across samples; 95% **confidence interval** $\approx$ estimate ± 1.96 SE
- **iid**: independent and identically distributed; most textbook formulas assume it
- **Stationary bootstrap** (Politis & Romano (1994)): resample blocks of consecutive days of random length (mean b) to keep the dependence in the data
- **Fisher z**: $\text{atanh}(\hat{\rho})$ approximately follows the Normal distribution with variance $1/(n-3)$ for iid pairs from a bivariate Normal distribution (Fisher (1915)); fat tails and dependence change this variance
- **MAD** (median absolute deviation): median of $|x - \text{median}|$; $1.4826 \times \text{MAD}$ estimates σ under normality
- **Hampel filter** (Hampel (1974)): flag x_t if $|x_t - m_t| > c \times 1.4826 \times \text{MAD}_t$, with m_t the median of a centred window and MAD_t the median of $|x_j - m_t|$ over the same window
- **Market model**: $r_{i,t} = a_i + b_i r_{m,t} + \varepsilon_{i,t}$
 - ▶ **abnormal return** $AR_{i,t} = r_{i,t} - \hat{a}_i - \hat{b}_i r_{m,t}$; **CAR**: cumulative abnormal return over an event window

Definitions Used Today (5/5): Asymptotics and Break Tests

- **Delta method:** if $\sqrt{T}(\hat{\theta} - \theta) \rightarrow N(0, \Sigma)$ and g is differentiable, then $\sqrt{T}(g(\hat{\theta}) - g(\theta)) \rightarrow N(0, \nabla g^\top \Sigma \nabla g)$
- **HAC standard errors** ([Newey & West \(1987\)](#)): OLS standard errors robust to heteroskedasticity and autocorrelation up to L lags, with Bartlett weights $1 - j/(L + 1)$
- **Known-date break test** ([Chow \(1960\)](#)): regress on a constant and a dummy $D_t = 1\{t \geq \tau\}$; test the dummy's coefficient
- **Unknown-date break test** ([Andrews \(1993\)](#)): the largest Wald statistic over all τ in the central 70% of the sample (15% trimming), compared with its own critical values
- **Break-date interval** ([Bai \(1997\)](#)): for one mean shift δ and the least-squares date $\hat{\tau}$, $\hat{\tau} \pm 11 \hat{\omega}^2 / \hat{\delta}^2$ observations, $\hat{\omega}^2 =$ long-run variance of the residuals; assumes the same error distribution in both regimes
- **Bootstrap p-value:** the share of resampled statistics at least as extreme as the observed one

Sources Used Today

- ▣ **FRED** (St. Louis Fed): public US macro, interest-rate and exchange-rate series
- ▣ **BNR**: publishes the official EUR/RON reference rate (fixing at about 13:00)
- ▣ **DefiLlama**: a public data platform for DeFi, including stablecoin supply
- ▣ **SPY**: S&P 500 ETF; **TLT**: iShares 20+ Year Treasury Bond ETF (long US government bonds)
- ▣ **BET**: the Bucharest Stock Exchange benchmark index; **BET-TR**: its version with reinvested dividends
- ▣ **Stablecoin supply**: the total value of USD-pegged tokens in circulation, in bn USD
- ▣ **S&P 500**: 500 large US companies; Euro Stoxx 50 and Nikkei 225: euro-area and Japanese blue-chip indices

One Function for All Data

```
URL = ('https://raw.githubusercontent.com/'
      'danpele/MFM/main/data/market/')

def fetch_daily(symbol, source='market',
               field='close'):
    if source == 'market':
        d = pd.read_csv(URL + symbol + '.csv',
                        index_col='date',
                        parse_dates=True)
        return d[field].dropna()
    if source == 'fred':
        u = ('https://fred.stlouisfed.org/graph/'
            'fredgraph.csv?id=' + symbol)
        return pd.read_csv(u, index_col=0,
                          parse_dates=True, na_values='.')\
            .iloc[:, 0].dropna()
    raise ValueError(source)

spx = fetch_daily('GSPC.INDX')
spy = fetch_daily('SPY.US', field='adjusted_close')
```

- One entry point: the rest of the code never knows where a file comes from
- `index_col='date'`: the same call as in the lecture code
- `close` for indices, FX, crypto, yields; `adjusted_close` for stocks and ETFs
- Public sources: FRED, BNR, DefiLlama

Cleaning Checklist

Calendars

- ▣ Equities: about 252 trading days a year; crypto: 365; spot gold: about 260 (weekend prints removed)
- ▣ Joint analysis: join the prices on common dates first, then compute returns
- ▣ Or aggregate to weekly returns (Friday to Friday)

▣ Exercises 6 and 6 extended apply this checklist to the EUR/RON market file

Values

- ▣ Adjusted vs. raw prices (dividends, splits)
- ▣ Weekend rows in FX or yield files
- ▣ Bad ticks: points far from a rolling median that reverse the next day
- ▣ Missing values: never fill returns with zeros silently

Exercises A1–A4 [Solved]: Warm-Up — Task

- **Data and assumptions:** the numbers below, no data file; $\ln$ is the natural logarithm; returns are daily unless stated otherwise
- **Tasks**
 - ▶ (a) A1: prices $P_0 = 100$, $P_1 = 110$, $P_2 = 99$. Compute the simple returns R_1, R_2 and the log returns r_1, r_2 , add each pair, and compare the sums with the two-period returns $P_2/P_0 - 1$ and $\ln(P_2/P_0)$.
 - ▶ (b) A2: prices 100, 120, 90, 110, 130, 100. Compute the running peak $M_t = \max_{s \leq t} P_s$, the drawdown $DD_t = P_t/M_t - 1$ on every day, and the maximum drawdown.
 - ▶ (c) A3: annualise a daily volatility of 1% with $\sqrt{252}$ for an equity index and with $\sqrt{365}$ for Bitcoin, assuming uncorrelated returns with constant variance.
 - ▶ (d) A4: the S&P 500 was multiplied by 3.72 between 2 Jan 2015 and 18 Sep 2026 (4 277 calendar days, $Y = 4\,277/365.25 = 11.71$ years). Compute the CAGR.
- **Report:** the four sets of numbers. **Interpretation:** why do log returns add up over time while simple returns do not, and which assumption does (c) need?

Exercises A1–A4 [Solved]: Warm-Up — Solution

- (a) Simple returns: $R_1 = 110/100 - 1 = 10\%$ and $R_2 = 99/110 - 1 = -10\%$; their sum is 0%, but the two-period return is $99/100 - 1 = -1\%$.
 - ▶ Log returns: $r_1 = \ln 1.10 = 9.53\%$ and $r_2 = \ln 0.90 = -10.54\%$; their sum is $-1.005\% = \ln 0.99 = \ln(P_2/P_0)$.
 - ▶ Log returns telescope, $\ln P_2 - \ln P_1 + \ln P_1 - \ln P_0$; simple returns compound, $(1 + R_1)(1 + R_2) - 1 = 1.1 \times 0.9 - 1 = -1\%$.
- (b) Running peak 100, 120, 120, 120, 130, 130; drawdowns 0, 0, $90/120 - 1 = -25\%$, $110/120 - 1 = -8.33\%$, 0, $100/130 - 1 = -23.08\%$.
 - ▶ The maximum drawdown is -25% , on day 3; the later fall to 100 is smaller because it is measured from the new peak of 130.
- (c) $\sqrt{252} = 15.87$, so $1\% \times 15.87 = 15.87\%$; $\sqrt{365} = 19.10$, so $1\% \times 19.10 = 19.10\%$.
 - ▶ The rule needs uncorrelated returns with constant variance; Exercise A6 shows what happens otherwise.
- (d) $\text{CAGR} = 3.72^{1/11.71} - 1 = \exp(\ln 3.72/11.71) - 1 = \exp(1.3137/11.71) - 1 = e^{0.1122} - 1 = 11.87\%$, the lecture's 11.9% a year.

Exercise A5 [Solved]: Volatility Drag — Task

- **Data and assumptions:** log returns r_t iid with mean $\mu_{\log}$ and variance σ^2 ; simple return $R_t = e^{r_t} - 1$ with mean μ_{arit}
 - ▶ Bitcoin, BTC-USD.CC (close), daily, 17 Sep 2014 – 18 Sep 2026, 365 observations a year: daily log mean 0.118%, daily standard deviation 3.49%
- **Tasks**
 - ▶ (a) Expand $\ln(1 + R)$ to second order around $R = 0$, take expectations, and show that $\mu_{geo} = E[\ln(1 + R)] \approx \mu_{arit} - \frac{1}{2}\sigma^2$.
 - ▶ (b) For Bitcoin, compute the annual log mean ($\times 365$), the annual $\sigma^2/2$, the implied arithmetic mean and the CAGR $= e^{\mu_{\log}} - 1$, and compare this CAGR with the one computed from the first and last prices.
 - ▶ (c) An asset has an annual arithmetic mean of 10% and an annual volatility of 60%. Compute its expected log growth and its CAGR.
- **Report:** the numbers of (b) and (c). **Interpretation:** can a positive arithmetic mean go together with falling wealth?

Exercise A5 [Solved]: Volatility Drag — Solution

- (a) $\ln(1 + R) = R - \frac{1}{2}R^2 + O(R^3)$; taking expectations, $E[\ln(1 + R)] \approx \mu_{arit} - \frac{1}{2}E[R^2]$, and $E[R^2] = \text{Var}(R) + \mu_{arit}^2$.
 - For small daily returns $\text{Var}(R) \approx \text{Var}(r) = \sigma^2$ and $\mu_{arit}^2 \approx 0$, so $\mu_{geo} \approx \mu_{arit} - \frac{1}{2}\sigma^2$: an approximation, not an identity.
- (b) Annual log mean: $0.118\% \times 365 = 43.1\%$.
 - Annual $\sigma^2/2$: $0.0349^2 \times 365/2 = 0.001218 \times 182.5 = 0.222$, i.e. 22.2 pp.
 - Implied arithmetic mean: $43.1\% + 22.2\% = 65.3\%$; the mean of daily simple returns times 365 gives 65.2%.
 - CAGR: $e^{0.431} - 1 = 53.9\%$, equal to the CAGR from the first and last prices (53.9%): in the sample, the sum of log returns is exactly $\ln(P_{\text{end}}/P_{\text{start}})$.
- (c) Using the approximation of (a) with annual moments: expected log growth $\approx 0.10 - 0.60^2/2 = -0.08$, long-run CAGR $\approx e^{-0.08} - 1 = -7.7\%$ (an approximation: if annual log returns follow the Normal distribution, the exact value is $\ln 1.10 - 0.18 = -0.085$, CAGR -8.1%).
- Interpretation: yes; with a volatility of 60% the drag $\sigma^2/2 = 18$ pp exceeds the 10% arithmetic mean, so the typical path loses money even though the average one-period return is positive.

Exercise A6 [Proposed]: When the $\sqrt{P}$ Rule Fails — Task

- **Model:** A1–A4 and A5 [Solved]
- **Data and assumptions:** daily log returns r_t , stationary, with variance σ^2 and autocorrelations ρ_j
 - ▶ EUR/RON BNR reference rate, 1 Jul 2005 – 18 Sep 2026: $\hat{\rho}_1 = 0.17$, volatility with $\sqrt{252}$ equal to 4.44%
 - ▶ EUR/RON market file `EURRON.FOREX`, 2015–2026, which contains bad ticks
- **Tasks**
 - ▶ (a) Write the variance of $\sum_{t=1}^k r_t$ as the sum of all covariances and derive $\text{Var} \left(\sum_{t=1}^k r_t \right) = k\sigma^2 \left[1 + 2 \sum_{j=1}^{k-1} \left(1 - \frac{j}{k} \right) \rho_j \right]$.
 - ▶ (b) For an AR(1), $\rho_j = \rho^j$, find the limit of the ratio of this variance to $k\sigma^2$ as $k \rightarrow \infty$.
 - ▶ (c) Correct the BNR volatility of 4.44% with the limit of (b) and with the exact formula for $k = 252$.
 - ▶ (d) The market file has $\hat{\rho}_1 = -0.57$: compute the AR(1) factor and say in which direction $\sqrt{252}$ is now wrong.
 - ▶ (e) Compute the BNR autocorrelations at lags 1–5 and the volatility from monthly returns times $\sqrt{12}$; does the AR(1) correction of (c) hold?
- **Report:** the factors and corrected volatilities. **Interpretation:** does positive autocorrelation hide risk or invent it?

Exercise A7 [Proposed]: Why Is the Weekly Correlation Lower than the Daily One? — Task

□ **Model:** A5 and Ex3 [Solved]

□ **Data and assumptions:** $x_t = \text{S\&P 500 (GSPC.INDX)}$, $y_t = \text{Bitcoin (BTC-USD.CC)}$, daily log returns on common days (prices joined first), 2022 – 18 Sep 2026, $n = 1\,182$

▶ no own autocorrelation; cross-covariances $\gamma(j) = \text{Cov}(x_t, y_{t-j})$

□ **Tasks**

▶ (a) For the 5-day sums X, Y , derive $\text{Cov}(X, Y) = 5\gamma(0) + \sum_{j=1}^4 (5-j)[\gamma(j) + \gamma(-j)]$, and show that the weekly correlation equals the daily one when $\gamma(j) = 0$ for $j \neq 0$.

▶ (b) Ex3 gives 0.42 daily and 0.30 weekly: state the sign the lagged covariances must have, measure the lagged term, and compute the implied 5-day correlation.

▶ (c) With $n = 246$ weeks and $\text{SE} \approx (1 - \hat{\rho}^2)/\sqrt{n}$, test at the 5% level whether 0.30 differs from 0.42.

▶ (d) Recompute the weekly correlation with weeks anchored on Monday, ..., Friday, each with a Fisher 95% interval.

□ **Report:** the lagged term, the implied correlation, the test and the five weekly values. **Interpretation:** is the gap a property of the markets or of the sampling? The [Epps \(1979\)](#) effect predicts the opposite sign.

Exercise A8 [Proposed]: the Standard Error of a Sharpe Ratio — Task

▣ **Model:** A5 and Ex1 Extended [Solved]

▣ **Data and assumptions:** T daily returns, iid, with mean μ , variance σ^2 , skewness γ_3 , kurtosis γ_4 (central moments μ_3, μ_4); $SR = \mu/\sigma$

▶ SPY (Ex1 Extended): $SR_a = 0.82$, $Y = 11.71$ years, $q = 251.4$ days a year, $T = 2\,944$, $\gamma_3 = -0.31$, $\gamma_4 = 17.1$

▣ **Tasks**

▶ (a) Using that $\sqrt{T}(\hat{\mu} - \mu, \hat{\sigma}^2 - \sigma^2)$ is asymptotically Normal with covariance $\begin{pmatrix} \sigma^2 & \mu_3 \\ \mu_3 & \mu_4 - \sigma^4 \end{pmatrix}$, apply the delta method

to $g(\mu, \sigma^2) = \mu/\sigma$ and derive the asymptotic variance of $\sqrt{T}(\widehat{SR} - SR)$.

▶ (b) Show that for Normal returns this variance is $1 + SR^2/2$ (Lo (2002)).

▶ (c) With $SR_a = \sqrt{q} SR_d$ and $T = qY$, show that $\text{Var}(\widehat{SR}_a) = (1 + SR_a^2/(2q))/Y \approx 1/Y$, whatever the frequency q .

▶ (d) Compute the SE of the SPY Sharpe ratio with (b) and with (a), and compare with $\sqrt{(1 + SR_a^2/2)/Y}$.

▣ **Report:** the variance formula and the three SEs. **Interpretation:** why does sampling more often not make the Sharpe ratio more precise?

Exercise A9 [Proposed]: Asynchronous Closes — Task

- **Model:** A5 and Ex3 [Solved]
- **Data and assumptions:** log prices $X(s) = W(s) + U(s)$ and $Y(s) = W(s) + V(s)$, with W, U, V independent Brownian motions, so that daily returns have correlation ρ
 - ▶ X closes at integer times t ; Y closes $\delta \in (0, 1)$ of a day earlier, at $t - \delta$
 - ▶ Data: Bitcoin (price of day t taken at 24:00 UTC, the end of day t) and the S&P 500 (close at 20:00 UTC in summer, 21:00 UTC in winter), 2022 – 18 Sep 2026
- **Tasks**
 - ▶ (a) Show that $\text{Corr}(x_t, y_t) = \rho(1 - \delta)$, $\text{Corr}(x_t, y_{t+1}) = \rho\delta$, and that all other cross-correlations are zero.
 - ▶ (b) Deduce that in this model the sum of the correlations at lags $-1, 0$ and $+1$ recovers ρ (Scholes & Williams (1977)).
 - ▶ (c) With $X = \text{Bitcoin}$ and $Y = \text{S\&P 500}$, compute δ in summer and in winter and the implied same-day and lead correlations for $\rho = 0.45$.
 - ▶ (d) Compare with the measured correlations: same day 0.42; S&P 500 with Bitcoin one day earlier +0.03; with Bitcoin one day later -0.05 .
- **Report:** δ , the predicted and measured correlations. **Interpretation:** is the asynchronous-close bias important for this pair?

Ex1 [Solved]: A Cross-Asset Dashboard — Task

- **Data:** daily, 2 Jan 2015 – 18 Sep 2026 ($Y = 11.71$ years), from data/market
 - ▶ SPY (SPY.US, adjusted close: the S&P 500 with dividends), TLT (TLT.US, adjusted close), gold (XAUUSD.FOREX, close, weekday quotes only), Bitcoin (BTC-USD.CC, close)
 - ▶ Euro Stoxx 50 (STOXX50E.INDX) and Nikkei 225 (N225.INDX), close, converted to USD with the FRED rates DEXUSEU and DEXJPUS of the same day
- **Tasks**
 - ▶ (a) Convert the two foreign indices to USD: $P^{USD} = P^{EUR} \times (\text{USD per EUR})$ and $P^{USD} = P^{JPY} / (\text{JPY per USD})$.
 - ▶ (b) Plot the growth of 1 USD, P_t/P_0 , for the six assets on a log scale.
 - ▶ (c) For each asset compute the growth multiple $P_{\text{end}}/P_{\text{start}}$, the CAGR = $\text{multiple}^{1/Y} - 1$, and the annualised volatility $\hat{\sigma}_d \sqrt{q}$, where $q = n/Y$ is the observed number of trading days per year.
 - ▶ (d) For gold, recompute the volatility with $q = 252$ and compare it with the GLD ETF (GLD.US, adjusted close).
- **Report:** a table with q , multiple, CAGR and volatility. **Interpretation:** how much does annualising gold with 252 instead of its actual frequency change its volatility?

Ex1 [Solved]: Solution (1/2) — Steps

- Step 1, SPY: 169.26 on 2 Jan 2015 and 761.69 on 18 Sep 2026, multiple $761.69/169.26 = 4.50$
 - ▶ $\text{CAGR} = 4.50^{1/11.71} - 1 = e^{1.504/11.71} - 1 = 13.7\%$
 - ▶ $n = 2944$ returns, $q = 2944/11.71 = 251.4$; $\hat{\sigma}_d = 1.11\%$, volatility $1.11\% \times \sqrt{251.4} = 17.6\%$
- Step 2, Bitcoin: 315.03 to 80901.46, multiple 256.8, $\text{CAGR} = 256.8^{1/11.71} - 1 = e^{5.548/11.71} - 1 = 60.6\%$
 - ▶ $q = 365.2$; $\hat{\sigma}_d = 3.49\%$, volatility $3.49\% \times \sqrt{365.2} = 66.7\%$; with $\sqrt{252}$ it would be 55.4%
- Step 3, gold (weekday quotes): 1189.18 to 4380.01, multiple 3.68, CAGR 11.8%
 - ▶ $q = 260.4$, $\hat{\sigma}_d = 1.01\%$, volatility 16.3%; with $\sqrt{252}$: 16.1%; GLD: 16.3%
- Step 4, the same procedure for Euro Stoxx 50, Nikkei 225 and TLT (next frame)

Ex1 [Solved]: Solution (2/2) — Table and Interpretation

- Results (q ; multiple; CAGR; volatility)
 - ▶ S&P 500 with dividends (SPY): 251; $\times 4.50$; 13.7%; 17.6%
 - ▶ Euro Stoxx 50 (USD): 256; $\times 1.90$; 5.6%; 20.7%
 - ▶ Nikkei 225 (USD): 245; $\times 2.85$; 9.4%; 21.3%
 - ▶ Gold: 260; $\times 3.68$; 11.8%; 16.3%
 - ▶ TLT: 251; $\times 0.88$; -1.0% ; 14.7%
 - ▶ Bitcoin: 365; $\times 256.8$; 60.6%; 66.7%
- Interpretation: once weekend prints are removed, gold has about 260 quotes a year; annualising with 252 instead moves its volatility by only 0.2 pp (16.3% vs 16.1%), and GLD confirms 16.3%.
- For Bitcoin the choice matters: 365 vs 252 changes the volatility by 11 pp.
- Long Treasuries (TLT) lost money in USD over the whole period; equities outside the US grew much less than the S&P 500.

Ex1 Extended [Solved]: How Reliable Are the Numbers in the Table? — Task

- **Data and assumptions:** the six series of Ex1, daily simple returns R_t , risk-free rate set to 0
- **Tasks**
 - ▶ (a) Compute the annualised Sharpe ratio $SR_a = \sqrt{q} \bar{R}/s_R$ of each asset.
 - ▶ (b) Compute its standard error for iid returns following the Normal distribution (Lo (2002)) from the daily Sharpe ratio $SR_d = \bar{R}/s_R$ of T daily returns: $SE(SR_a) = \sqrt{q(1 + SR_d^2/2)/T}$.
 - ▶ (c) Compute the SE corrected for skewness γ_3 and kurtosis γ_4 (Opdyke (2007)): $Var(SR_d) \approx [1 + \frac{1}{2}SR_d^2 - \gamma_3SR_d + \frac{\gamma_4-3}{4}SR_d^2]/T$, times q .
 - ▶ (d) Run a stationary bootstrap (Politis & Romano (1994)) with mean block 20 days and 2000 replications, and report the 95% percentile intervals for the CAGR and the Sharpe ratio.
 - ▶ (e) Repeat (d) with mean blocks of 5 and 60 days (1 000 replications each).
 - ▶ (f) On the common days of SPY and gold, bootstrap (block 20, 2000 replications) the difference of their Sharpe ratios and report its 95% interval.
 - ▶ (g) Repeat (f) for the 10 pairs of the five non-crypto assets, with a Bonferroni correction: each test at $5\%/10 = 0.5\%$.
- **Report:** a table with SR, the two SEs and the bootstrap intervals. **Interpretation:** which differences between assets survive once uncertainty is added? Is the iid formula too optimistic for Bitcoin? What does the dependence on the block length say about the data?

Ex1 Extended [Solved]: Solution (1/2) — Worked Example, SPY

- (a) $SR_a = 0.82$ with $q = 251.4$, so $SR_d = 0.82/\sqrt{251.4} = 0.052$
- (b) $SE = \sqrt{251.4 \times (1 + 0.052^2/2)/2944} = \sqrt{0.0855} = 0.29$ ($\approx 1/\sqrt{11.71}$); 95% interval $0.82 \pm 1.96 \times 0.29 = [0.25; 1.39]$
- (c) With $\gamma_3 = -0.31$ and $\gamma_4 = 17.1$ the bracket is $1 + 0.0013 + 0.016 + 0.009 = 1.027$: $SE = 0.30$
- (d) Bootstrap procedure
 - ▶ draw a start day at random; continue the block with probability $1 - 1/20$ or jump to a new random day with probability $1/20$, until T days are drawn
 - ▶ recompute the CAGR and SR on the resampled series; repeat 2 000 times; take the 2.5% and 97.5% quantiles
 - ▶ SPY: SR [0.32; 1.40], CAGR [4.4%; 23.7%]
- (e) Blocks of 5 and 60 days: [0.29; 1.39] and [0.37; 1.35]
- (f) SPY minus gold on 2944 common days: +0.06, 95% interval $[-0.65; +0.81]$
- (g) SPY minus Euro Stoxx 50: +0.46, [0.09; 0.85], $p = 0.012$; the smallest of the 10 bootstrap p-values is 0.008 (gold minus TLT), above 0.005

Ex1 Extended [Solved]: Solution (2/2) — Table and Interpretation

- ▣ Sharpe ratio (SE Lo; SE Opdyke; bootstrap 95%, block 20)
 - ▶ S&P 500: 0.82 (0.29; 0.30; [0.32, 1.40]); gold: 0.77 (0.29; 0.30; [0.23, 1.29])
 - ▶ Nikkei: 0.53 (0.29; 0.29; [0.07, 1.02]); Euro Stoxx: 0.37 (0.29; 0.29; [-0.17, 0.91])
 - ▶ TLT: 0.00 (0.29; 0.29; [-0.52, 0.51]); Bitcoin: 1.05 (0.29; 0.29; [0.44, 1.66])
- ▣ Overlapping intervals are not a test; the bootstrapped differences are: S&P 500 and gold cannot be distinguished, S&P 500 minus Euro Stoxx 50 is significant on its own, and no pair survives the Bonferroni correction: 11.7 years establish no robust Sharpe ranking.
- ▣ The CAGR is even less precise: S&P 500 [4.4%, 23.7%], Bitcoin [7%, 141%].
- ▣ With daily data every iid SE is $\approx 1/\sqrt{11.71} = 0.29$; the daily SR is so small that fat tails barely move Opdyke's SE; Bitcoin's bootstrap width, $1.22/3.92 = 0.31$, is slightly above the iid value.
- ▣ The block length changes the intervals little: serial dependence in daily returns is weak.

Ex2 [Proposed]: Drawdowns and Recovery Time — Task

- **Model:** A1–A4 and Ex1 [Solved]
- **Data:** S&P 500 (GSPC.INDX, close), 3 Jan 2000 – 18 Sep 2026; Bitcoin (BTC-USD.CC, close), 17 Sep 2014 – 18 Sep 2026; daily
- **Convention:** a spell below a previous peak starts on the peak date and ends on the first close at or above that peak
- **Tasks**
 - ▶ (a) Compute the drawdown series $DD_t = P_t / \max_{s \leq t} P_s - 1$ of both assets and plot it.
 - ▶ (b) Find the maximum drawdown and its date, with the peak and trough prices.
 - ▶ (c) Find the longest spell below a previous peak, with its start and end dates and its length in calendar days.
 - ▶ (d) Report the current drawdown on 18 Sep 2026.
- **Report:** the numbers of (b)–(d) for both assets. **Interpretation:** Bitcoin's peak date depends on the daily close convention of the source (snapshot at 24:00 UTC, the end of the UTC day, or another time zone); which of your results are sensitive to it?

Ex2 Extended [Proposed]: Is the 2009 Drawdown Extreme? — Task

- **Model:** Ex1 Extended [Solved]
- **Data and assumptions:** S&P 500 daily log returns, 3 Jan 2000 – 18 Sep 2026, $n = 6\,717$; observed maximum drawdown -56.8% and longest spell under water 1 803 trading days (Ex2)
- **Tasks**
 - ▶ (a) Generate 5 000 return paths of length n with the iid bootstrap and 5 000 with the stationary bootstrap (mean block 20 days).
 - ▶ (b) For every path compute the maximum drawdown and the longest spell under water; report the median, the 5% quantile and the bootstrap p-values $P(\text{MDD}^* \leq -56.8\%)$ and $P(\text{spell}^* \geq 1\,803)$.
 - ▶ (c) Compute the lag-1 autocorrelation $\hat{\rho}_1$ and the variance ratio $\widehat{\text{VR}}(20) = \text{Var}(20\text{-day return}) / (20 \hat{\sigma}^2)$ of the daily returns.
- **Report:** the two distributions and the p-values. **Interpretation:** is -56.8% extreme at the 5% level for this return distribution? Which bootstrap gives the fatter drawdown tail, and why?

Ex3 [Solved]: Correlations and Calendars — Task

- **Data:** S&P 500 (GSPC.INDX), Bitcoin (BTC-USD.CC) and TLT (TLT.US, adjusted close), daily log returns
 - ▶ Bitcoin – S&P 500: 2022 – 18 Sep 2026; stock–bond (S&P 500 – TLT): 2010–2020 and 2022 – 18 Sep 2026
- **Tasks**
 - ▶ (a) Compute the Bitcoin – S&P 500 correlation over the whole period on common days: join the prices on the days both trade, then take returns.
 - ▶ (b) Recompute it with the S&P 500 forward-filled on weekends (zero weekend returns) and with weekly returns, Friday to Friday.
 - ▶ (c) Compute the mean of the 252-day rolling correlation (the lecture's estimator) over the same period.
 - ▶ (d) Compute the stock–bond correlation in both periods with both estimators, whole period and mean rolling.
- **Report:** the correlations and sample sizes. **Interpretation:** why do the two estimators differ, and which one answers “what was the correlation in 2022–2026”?

Ex3 [Solved]: Solution

- (a) Common days: $n = 1\,182$ pairs of returns; correlation 0.42.
- (b) Forward-filled: 0.39, because the zero weekend returns of the S&P 500 are paired with Bitcoin's weekend moves; weekly (246 weeks): 0.30.
- (c) Mean of the 252-day rolling correlation: 0.38 (the lecture's value).
- (d) Stock–bond, whole period: -0.47 (2010–2020) and $+0.11$ (2022–2026); mean rolling: -0.43 and $+0.08$.
- Interpretation
 - ▶ the whole-period correlation uses all pairs at once and answers “what was the correlation in 2022–2026”;
 - ▶ the mean of rolling correlations weights every one-year window equally, while the whole-period correlation gives more weight to high-volatility windows, where the correlation is larger in absolute value;
 - ▶ the calendar choice alone moves the Bitcoin estimate between 0.30 and 0.42.

Ex3 Extended [Proposed]: Has the Stock–Bond Correlation Really Changed? — Task

□ **Model:** Ex1 Extended [Solved]

□ **Data:** Ex3; S&P 500 – TLT: $\hat{\rho}_1 = -0.47$ (2010–2020, $n_1 = 2\,769$) and $\hat{\rho}_2 = +0.11$ (2022–2026, $n_2 = 1\,182$)

□ **Tasks**

▶ (a) Test $H_0 : \rho_1 = \rho_2$ at the 5% level with the Fisher z test, valid for iid pairs from a bivariate Normal distribution:

$$z = \frac{\operatorname{atanh} \hat{\rho}_2 - \operatorname{atanh} \hat{\rho}_1}{\sqrt{1/(n_1 - 3) + 1/(n_2 - 3)}}.$$

▶ (b) Run a stationary bootstrap separately in each period (mean block 20 days, 2 000 replications) and report the 95% interval of $\hat{\rho}_2 - \hat{\rho}_1$.

▶ (c) Compare the width of the bootstrap interval with the iid interval $\pm 1.96 \sqrt{(1 - \hat{\rho}_1^2)^2/(n_1 - 3) + (1 - \hat{\rho}_2^2)^2/(n_2 - 3)}$.

▶ (d) For Bitcoin – S&P 500 (2022–2026), give Fisher and bootstrap 95% intervals for the daily (block 20) and the weekly (block 4) correlation, and a joint bootstrap interval for their difference: resample whole weeks (mean block 4) and recompute both correlations from each resample.

□ **Report:** z , p -value, both intervals and their width ratio. **Interpretation:** why is the bootstrap interval wider? Does the stock–bond change remain significant, and does the daily–weekly gap of A7? Which economic event explains the change of sign?

Ex4 [Proposed]: Stablecoin Growth — Task

- **Model:** A1–A4 and Ex1 [Solved]
- **Data:** total supply of USD-pegged stablecoins, daily, in bn USD, from [DefiLlama](#), 31 Dec 2019 – 18 Sep 2026 (the 2019 year-end value is the base of the 2020 growth)
- **Tasks**
 - ▶ (a) Extract the year-end values and compute the year-on-year growth $X_{\text{year end}}/X_{\text{previous year end}} - 1$.
 - ▶ (b) Find the 2022 peak and the following trough up to the end of 2023, with exact dates, and compute the peak-to-trough decline.
 - ▶ (c) Compute the compound annual growth rate from 1 Jan 2020 to 18 Sep 2026.
- **Report:** the table of (a) and the numbers of (b)–(c). **Interpretation:** supply is a stock, not a price; what does a fall in supply mean economically, and would the same fall in a stablecoin's price mean the same?

Ex4 Extended [Proposed]: Did the Terra Collapse Change the Growth Regime? — Task

- **Model:** Ex7 [Solved] (regression with an event dummy)
- **Data:** $g_t = \Delta \ln(\text{supply}_t)$, daily, 1 Jan 2020 – 18 Sep 2026 ($n = 2\,452$); Terra collapse: 9 May 2022
- **Tasks**
 - ▶ (a) Known date ([Chow \(1960\)](#)): estimate $g_t = a + b1\{t \geq 9 \text{ May } 2022\} + e_t$ with HAC standard errors ([Newey & West \(1987\)](#), $L = 8$ lags) and test $b = 0$ at the 5% level.
 - ▶ (b) Unknown date ([Andrews \(1993\)](#)): compute the HAC Wald statistic for every break date in the central 70% of the sample (15% trimming), take the maximum, and compare it with the 5% critical value 8.85.
 - ▶ (c) Estimate the break date by least squares (minimum sum of squared residuals over the same dates) and its 95% interval ([Bai \(1997\)](#)): $\hat{\tau} \pm 11 \hat{\omega}^2 / \hat{b}^2$ days, $\hat{\omega}^2$ the Bartlett long-run variance of the residuals ($L = 8$); this interval assumes one break and the same error distribution in both regimes: check it.
- **Report:** $\hat{a}$, $\hat{b}$ (annualised $\times 365$), t , the sup-Wald statistic and date, the interval. **Interpretation:** did the slowdown start with Terra? Is the break in the level or in the drift of supply?

Ex5 [Proposed]: BVB in RON and in EUR — Task

- **Model:** Ex1 (currency conversion) [Solved]
- **Data:** BET-TR (BETTR.INDX, close) and the official EUR/RON reference rate of the BNR, daily, 5 Jan 2015 – 18 Sep 2026 ($Y = 11.70$)
- **Tasks**
 - ▶ (a) Convert BET-TR to euro: $P_t^{EUR} = P_t^{RON} / S_t$, with S_t the EUR/RON rate of the same day.
 - ▶ (b) Compute the CAGR and the annualised volatility (252 days) in RON and in EUR.
 - ▶ (c) Compute the correlation of daily BET-TR log returns with the daily log changes of EUR/RON.
- **Report:** the two CAGRs, the two volatilities and the correlation. **Interpretation:** for a euro-based investor, is the leu's depreciation a cost or a hedge? What would change your answer?

Ex5 Extended [Proposed]: Is the Currency Effect Significant? — Task

- **Model:** Ex1 Extended and Ex7 [Solved]
- **Data:** Ex5; log returns $r_t^{EUR} = r_t^{RON} - \Delta s_t$, with $s_t = \ln(\text{EUR}/\text{RON})$; $q = 250$ observations a year
- **Tasks**
 - ▶ (a) Show that $\text{Var}(r^{EUR}) = \text{Var}(r^{RON}) + \text{Var}(\Delta s) - 2 \text{Cov}(r^{RON}, \Delta s)$ and compute each term annualised ($\times q$).
 - ▶ (b) Test $\text{Cov}(r^{RON}, \Delta s) = 0$ at 5%: regress r_t^{RON} on Δs_t with HAC standard errors ([Newey & West \(1987\)](#), $L = 8$), and compute a stationary-bootstrap 95% interval (block 20, 2 000 replications) for the correlation.
 - ▶ (c) Estimate the FX drag, the mean of Δs_t times q , with its HAC standard error, t statistic and 95% interval.
- **Report:** the decomposition, the slope with its t , the intervals. **Interpretation:** is the 1.6 pp gap between the two CAGRs distinguishable from zero? Would a euro investor hedge the leu?

Ex6 [Solved]: Finding Bad Ticks — Task

- **Data:** EUR/RON market file EURRON.FOREX (close), daily, 2015 – 18 Sep 2026, 3 115 rows; BNR reference rate for comparison
- **Tasks**
 - ▶ (a) Drop the weekend rows.
 - ▶ (b) Flag every weekday point more than 0.05 RON away from the 11-day centred rolling median, and remove the flagged points.
 - ▶ (c) Compute the annualised volatility of the raw series, of the cleaned series and of the BNR rate, each with its observed number of quotes per year (FX: not 252).
 - ▶ (d) Compare the cleaned series with the BNR rate: correlation of daily changes, lag-1 autocorrelations, days quoted in the file without a BNR fixing.
- **Report:** the counts, the largest deviation and the three volatilities. **Interpretation:** the cleaned series still has a volatility of about 2.6% against 2.1% for BNR; why does a gap remain after the errors are removed?

Ex6 [Solved]: Solution

- (a) 58 weekend rows are removed.
- (b) 11 weekday points are flagged; the largest deviation is 0.84 RON, on 13 Aug 2025 (a quote about 17% away from the market).
- (c) Volatility, annualised with the observed number of quotes per year: 12.4% raw, 2.6% cleaned, 2.1% BNR: eleven bad ticks multiply the volatility by almost five.
- (d) The remaining gap
 - ▶ the file quotes a market price at another time than the 13:00 fixing: daily changes, on the common days of the two price series, correlate only 0.40 with BNR's;
 - ▶ lag-1 autocorrelation -0.11 in the file vs $+0.08$ at BNR, each on its own calendar: quote noise that reverses the next day;
 - ▶ 116 days are quoted in the file but have no BNR fixing (Romanian holidays), and deviations below 0.05 RON stay in the series.
- Interpretation: use the official BNR rate for EUR/RON; a threshold rule removes gross errors but not quote noise.

Ex6 Extended [Proposed]: A Calibrated Threshold — Task

- **Model:** Ex6 [Solved]
- **Data:** BNR EUR/RON rate 2015–2026 (treated as clean) and the market file of Ex6, weekdays only
- **Method:** Hampel filter ([Hampel \(1974\)](#)): flag day t if $|x_t - m_t| > c \cdot 1.4826 \cdot \text{MAD}_t$, with m_t the median of the centred 11-day window and MAD_t the median of $|x_j - m_t|$ over the same window (truncated at the ends of the sample)
- **Tasks**
 - ▶ (a) On the BNR series, compute the share of flagged days for $c = 3, 5, 10, 20$, and find the smallest c on the grid $1, 1.5, \dots, 59.5$ that flags at most 0.1% of the days (an in-sample false-alarm rate).
 - ▶ (b) Modify the rule with a floor, threshold $\max(c \cdot 1.4826 \cdot \text{MAD}_t, \text{floor})$: for floors of 0.01, 0.02 and 0.03 RON, find the smallest c on the same grid.
 - ▶ (c) Apply the calibrated rule without a floor and with the 0.01 RON floor to the market file: count the flagged days and those also flagged by the 0.05 RON rule of Ex6.
 - ▶ (d) Compute the volatility after cleaning and compare it with 2.1% (BNR).
- **Report:** the calibration table, c^* , the counts and the volatility. **Interpretation:** why MAD and not the standard deviation, and where does 1.4826 come from? What happens to the filter when MAD_t is almost zero?

Ex7 [Solved]: Event Study, the Bank Tax of December 2018 — Task

- **Data:** Banca Transilvania (TLV.R0, adjusted close) and BET (BET, close) as the market, daily log returns
- **Event:** the bank tax was announced on the evening of 18 Dec 2018; event day 0 = 19 Dec 2018
- **Method:** MacKinlay (1997); Campbell, Lo & MacKinlay (1997), Ch. 4

□ Tasks

- ▶ (a) Estimate the market model $r_{i,t} = a_i + b_i r_{m,t} + \varepsilon_{i,t}$ by OLS on the window $[-250, -11]$ trading days; report $\hat{b}_i$ and $\hat{\sigma}_{\varepsilon,i}$.
- ▶ (b) Compute the abnormal returns $AR_{i,t} = r_{i,t} - \hat{a}_i - \hat{b}_i r_{m,t}$ for days 0 to +5 and their sum $CAR_i[0, +5]$.
- ▶ (c) Test $AR_0 = 0$ and $CAR = 0$ at the 5% level with the prediction-error variances, which include the estimation error of $\hat{a}, \hat{b}$ ($T = 240$, $L = 6$, $S_{mm} = \sum (r_{m,t} - \bar{r}_m)^2$ on the estimation window; Student t with $T - 2$ degrees of freedom):

$$\text{Var}(AR_0) = \hat{\sigma}_{\varepsilon}^2 \left[1 + \frac{1}{T} + \frac{(r_{m,0} - \bar{r}_m)^2}{S_{mm}} \right], \quad \text{Var}(CAR) = \hat{\sigma}_{\varepsilon}^2 \left[L + \frac{L^2}{T} + \frac{(\sum_{\tau=0}^5 r_{m,\tau} - L\bar{r}_m)^2}{S_{mm}} \right]$$

- **Report:** $\hat{b}$, $\hat{\sigma}_{\varepsilon}$, the ARs, CAR, t and p . **Interpretation:** BET contains both big banks with large weights; what does this do to $\hat{b}$ and to AR? The t test assumes no change of variance in the event window; is that plausible here? Was there a reversal, and what does it say about market efficiency (Chapter 2)?

Ex7 [Solved]: Solution (1/2) — Computation

- (a) OLS on days $[-250, -11]$: $\hat{b} = 1.18$, $\hat{\sigma}_\varepsilon = 0.83\%$.
- (b) Day 0: $r_i = -22.2\%$ (log), market $r_m = -11.9\%$
 - ▶ $AR_0 = r_i - \hat{a} - \hat{b} r_m = -22.2\% - \hat{a} + 1.18 \times 11.9\% = -8.3\%$
 - ▶ days +1 to +5: $+3.2\%, +0.6\%, +2.5\%, +0.1\%, +0.9\%$
 - ▶ $CAR[0, +5] = -8.3 + 3.2 + 0.6 + 2.5 + 0.1 + 0.9 = -1.1\%$
- (c) Day 0: $S_{mm} = 0.0128$, $(r_{m,0} - \bar{r}_m)^2 / S_{mm} = 0.1195^2 / 0.0128 = 1.11$: $SE = 0.83\% \times \sqrt{2.12} = 1.21\%$
 - ▶ $t = -8.3 / 1.21 = -6.9$, $p < 0.001$: reject $AR_0 = 0$; with $\hat{\sigma}_\varepsilon$ alone t would be -10.0
- (c) CAR: $SE = 0.83\% \times \sqrt{6 + 0.15 + 1.33} = 2.27\%$, $t = -1.1 / 2.27 = -0.48$, $p = 0.63$: do not reject $CAR = 0$ at 5%.

Ex7 [Solved]: Solution (2/2) — Interpretation

- Day 0 is a huge surprise, but the abnormal returns of days 1–3 reverse most of it, so the six-day CAR is not significant.
- The market itself fell 11.9% because of the same news: BET contains the banks, the market model removes most of the shock, and AR understates it.
 - ▶ a better benchmark is BET without banks or a regional index
- The market's −11.9% day, far from its usual range, more than doubles the variance of AR_0 : the estimated $\hat{b}$ matters most on such days.
- Event days are also more volatile than the estimation window (event-induced variance), which even the full formula ignores; the [Boehmer, Musumeci & Poulsen \(1991\)](#) test divides the cross-sectional mean of standardised ARs by their cross-sectional standard error, so it needs many firms, not one bank.
- The reversal is consistent with an overreaction followed by a correction, or with the policy being partly withdrawn; Ex8 asks how to tell them apart.

Ex8 [Proposed]: The Same Event for BRD — Task

- **Model:** Ex7 [Solved]
- **Data:** BRD (BRD.R0, adjusted close) and BET, daily log returns; event day 0 = 19 Dec 2018
- **Tasks**
 - ▶ (a) Estimate the market model on $[-250, -11]$ and report $\hat{b}$ and $\hat{\sigma}_\varepsilon$.
 - ▶ (b) Compute AR_0 with its t , the ARs of days +1 to +5, and $CAR[0, +5]$ with t and p , as in Ex7.
 - ▶ (c) Compare the two banks: which one was hit harder relative to its normal volatility?
 - ▶ (d) Explain how the cross-sectional test of [Boehmer, Musumeci & Poulsen \(1991\)](#) standardises the abnormal returns, and why it cannot simply be applied to one or two banks hit by the same event.
- **Report:** the numbers of (a)–(b) and the comparison. **Interpretation:** for both banks, is the reversal a sign of overreaction or of a partly withdrawn policy? What evidence would separate the two?

C1: Did Spot ETFs Change Bitcoin's Correlation with Equities? — Task

- ▣ **Question:** spot Bitcoin ETFs were approved in the US on 10 Jan 2024; did the Bitcoin – S&P 500 correlation rise *because* of them, or had it risen already since 2022? There is no single method: propose one and defend it.
- ▣ **Data:** Bitcoin (BTC-USD.CC) and S&P 500 (GSPC.INDX), daily log returns on common days, 2018 – 18 Sep 2026; any other series of data/market as a control
- ▣ **Tasks**
 - ▶ (a) State a precise hypothesis: which parameter would have changed, from when, and in which direction.
 - ▶ (b) Choose between a known-date test ([Chow \(1960\)](#)) and an unknown-date test ([Andrews \(1993\)](#)), and justify the choice.
 - ▶ (c) Estimate the correlation in sub-periods with stationary-bootstrap 95% intervals, and test the change at the ETF date at the 5% level.
 - ▶ (d) Check robustness: calendar (common days, weekly), window length, bootstrap block length.
 - ▶ (e) Build a counterfactual: what happened in the same period to the correlation of assets without an ETF event?
- ▣ **Deliverable:** a Quantlet folder in your team repository (code, Metainfo, one chart) and one slide with the conclusion and its main limitation; for the curious, the dynamic DCC correlation of Chapter 6

C1: Reference Analysis

- ▣ *Solution discussed in class*
- ▣ Your Quantlet is compared with a reference analysis after the deadline.

C2 [Proposed]: Critique an AI Answer (1/2)

- **The prompt** a student sent to an AI (Artificial Intelligence) assistant: “Using the daily files in data/market, compute the annualised mean, the annualised volatility and the Sharpe ratio ($r_f = 0$) of SPY and Bitcoin from daily simple returns”
- **The answer:** the code below, followed by the conclusion

```
import numpy as np, pandas as pd
URL = 'https://raw.githubusercontent.com/danpele/MFM/main/data/market/'
def load(sym, field='close'):
    return pd.read_csv(URL + sym + '.csv', index_col='date',
                       parse_dates=True)[field].dropna()
spy, btc = load('SPY.US'), load('BTC-USD.CC')
A = 252                                     # trading days per year
for name, p in [('SPY', spy), ('BTC', btc)]:
    r = p.pct_change().dropna()             # daily simple returns
    mu = r.mean() * A                       # annualised mean
    vol = r.std() * A                       # annualised volatility
    print(name, mu, vol, mu / vol)          # Sharpe ratio, rf = 0
```

- **Conclusion of the AI:** “Sharpe ratios: SPY 0.03, Bitcoin 0.07. Both assets barely reward their risk.”

C2 [Proposed]: Critique an AI Answer (2/2) — Task

- ▣ **Model:** A1–A4 and Ex1 [Solved]; the definitions of today
- ▣ **Data and assumptions:** the prompt and the code of the previous frame; full files SPY.US and BTC-USD.CC; daily simple returns; risk-free rate 0
- ▣ **Tasks**
 - ▶ (a) Find the errors in the answer: there are two or three, and the rest of the answer is correct.
 - ▶ (b) For each error, say how you would detect it: a test, a published number, the lecture or the library documentation.
 - ▶ (c) Write the corrected code and report the annualised mean, the annualised volatility and the Sharpe ratio of SPY and Bitcoin.
 - ▶ (d) Log each error as in AI_ERRORS.md: what was asked, what the AI answered, what was wrong, how you found out, the correct version.
- ▣ **Report:** the list of errors and the corrected table. **Interpretation:** does the AI's conclusion “both assets barely reward their risk” survive?
- ▣ AI code usually runs without an error message: the mistakes are in the finance, not in the syntax
- ▣ Course rule: you may use AI, declare it in AI_USE.md and log at least three errors you caught in AI_ERRORS.md

C3 [Proposed]: Passive Investing and Demand Elasticity — Task (1/2)

- **Context:** the lecture case study (Haddad, Huebner & Loualiche (2025)) extends the demand system of Koijen & Yogo (2019) with a degree of strategic response χ and finds that one-third of a fall in the share of active investors passes through to the aggregate demand elasticity $\mathcal{E}_{agg}$ of single stocks
 - ▶ the 13F holdings are not in the course data, so part (a) uses the Financial Accounts of the United States and parts (c)–(e) use a simulated cross-section built with the paper's equations
- **Model:** Ex1 Extended [Solved] for reporting uncertainty across replications; the lecture case study for eq. (26)–(28), footnotes 53 and 55, and Table 4 of the paper
- **Data:** B0GZ1LM563064100Q (exchange-traded funds; corporate equities) and B0GZ1LM893064105Q (all sectors; corporate equities), from FRED, quarterly market values, 2001 Q1 to the latest quarter
- **Tasks**
 - ▶ (a) Compute the ETF share of corporate equities in each quarter, report it for 2001 Q1, 2020 Q4 and the latest quarter, and apply the rule of Section V.A of the paper (81% active investors at the start, pass-through 0.33) to obtain the implied fall in the active share and in $\mathcal{E}_{agg}$ over 2001–2020 and up to the latest quarter.
 - ▶ (b) Repeat (a) with the lowest and the highest pass-through of footnote 55 (31% for $\chi = 3.27$, 39% for $\chi = 2.30$), and state which part of the paper's narrow measure of passive investing the ETF share leaves out.
 - ▶ (c) Simulate 2000 stocks, each held by 40 active investors and by passive investors: draw the active share $|Active_k|$ uniform on (0.55, 0.85), split it among the 40 investors with flat Dirichlet weights, draw each own elasticity $\tilde{\mathcal{E}}_{ik}$ log-Normal with median 1.6 and log standard deviation 0.6, and compute $\mathcal{E}_{agg,k}$ from footnote 53 with $\chi = 2.97$.

C3 [Proposed]: Passive Investing and Demand Elasticity — Task (2/2)

▣ Tasks (continued)

- ▶ (d) In each stock, let a fraction f_k , uniform on $(0, 0.2)$, of the active holdings turn passive under three rules: every active investor in proportion (the representative case of Section V.A), the most elastic investors first, and the least elastic first; recompute $\mathcal{E}_{agg,k}$ and regress the change in $\log \mathcal{E}_{agg,k}$ on the change in $\log |Active_k|$ with a constant, as in column 1 of Table 4.
- ▶ (e) Repeat (c)–(d) for 10 random seeds and for $\chi = 0$, and compare each slope with the average theoretical pass-through $1/(1 + \chi|Active_k|)$ of eq. (27).
- ▣ **Report:** the ETF shares and the implied changes of (a)–(b); a table of slopes (mean, minimum and maximum over the seeds) for the three rules and the two values of χ , next to the theoretical pass-through
- ▣ **Interpretation:** why is the slope close to one-third only when the switchers are representative, and how should the 11% of the paper be read if the investors who turned passive were the most elastic ones?
- ▣ **Deliverable:** a Quantlet folder in your team repository (code, Metainfo, one table); a possible start for a team project on passive investing and price formation

Forming Your Team

Steps

- ▣ Form teams of 3–4 students
- ▣ Create the team repository on GitHub and add the instructors as collaborators
- ▣ Register the team and a tentative topic by the date announced in class
- ▣ The project is 70% of the grade: start the repository today and commit every week

A good first topic

- ▣ One market, one question, real data
- ▣ Example: do Romanian blue chips move with European banks or with energy prices?
- ▣ Example: Part C1 above, extended to Ethereum

Take-Home Assignment

Before the next seminar

- ▣ Run the seminar notebook end to end in Colab and fill in the empty cells
- ▣ Add one asset of your choice from `data/market` to `Ex1` extended, with its Sharpe interval
- ▣ Push the result to your team repository as your first Quantlet folder
- ▣ Answer the Chapter 0 quiz on the course website
- ▣ Log every AI error you catch, in this seminar or in the project, in `AI_ERRORS.md`

Further Study on Quantinar

Recommended courses

-  Statistics of Financial Markets
-  Cryptos, NFTs, Digital Assets
-  Introduction to Blockchain and Cryptocurrencies
-  Digital Transformation on Finance

References (1/3)

- ▣ Andrews, D.W.K. (1993). Tests for parameter instability and structural change with unknown change point. *Econometrica*, 61(4), 821–856.
- ▣ Bai, J. (1997). Estimation of a change point in multiple regression models. *Review of Economics and Statistics*, 79(4), 551–563.
- ▣ Boehmer, E., Musumeci, J., Poulsen, A.B. (1991). Event-study methodology under conditions of event-induced variance. *Journal of Financial Economics*, 30(2), 253–272.
- ▣ Campbell, J.Y., Lo, A.W., MacKinlay, A.C. (1997). *The Econometrics of Financial Markets*, Chapter 4. Princeton University Press.
- ▣ Chow, G.C. (1960). Tests of equality between sets of coefficients in two linear regressions. *Econometrica*, 28(3), 591–605.
- ▣ Epps, T.W. (1979). Comovements in stock prices in the very short run. *Journal of the American Statistical Association*, 74, 291–298.
- ▣ Fisher, R.A. (1915). Frequency distribution of the values of the correlation coefficient in samples from an indefinitely large population. *Biometrika*, 10(4), 507–521.
- ▣ Franke, J., Härdle, W.K., Hafner, C.M. (2019). *Statistics of Financial Markets: An Introduction* (5th ed.). Springer.

References (2/3)

- ▣ Haddad, V., Huebner, P., Loualiche, E. (2025). [How competitive is the stock market? Theory, evidence from portfolios, and implications for the rise of passive investing.](#) *American Economic Review*, 115(3), 975–1018.
- ▣ Hampel, F.R. (1974). [The influence curve and its role in robust estimation.](#) *Journal of the American Statistical Association*, 69(346), 383–393.
- ▣ Härdle, W.K., Harvey, C.R., Reule, R.C.G. (2020). [Understanding cryptocurrencies.](#) *Journal of Financial Econometrics*, 18(2), 181–208.
- ▣ Koijen, R.S.J., Yogo, M. (2019). [A demand system approach to asset pricing.](#) *Journal of Political Economy*, 127(4), 1475–1515.
- ▣ Lo, A.W. (2002). [The statistics of Sharpe ratios.](#) *Financial Analysts Journal*, 58(4), 36–52.
- ▣ MacKinlay, A.C. (1997). [Event studies in economics and finance.](#) *Journal of Economic Literature*, 35(1), 13–39.
- ▣ Newey, W.K., West, K.D. (1987). [A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix.](#) *Econometrica*, 55(3), 703–708.

References (3/3)

- ▣ Opdyke, J.D. (2007). [Comparing Sharpe ratios: so where are the p-values?](#) *Journal of Asset Management*, 8(5), 308–336.
- ▣ Politis, D.N., Romano, J.P. (1994). [The stationary bootstrap](#). *Journal of the American Statistical Association*, 89(428), 1303–1313.
- ▣ Scholes, M., Williams, J. (1977). [Estimating betas from nonsynchronous data](#). *Journal of Financial Economics*, 5(3), 309–327.
- ▣ Public data: [FRED](#), [BNR reference rates](#), [DefiLlama](#)
- ▣ Code and data: github.com/danpele/MFM